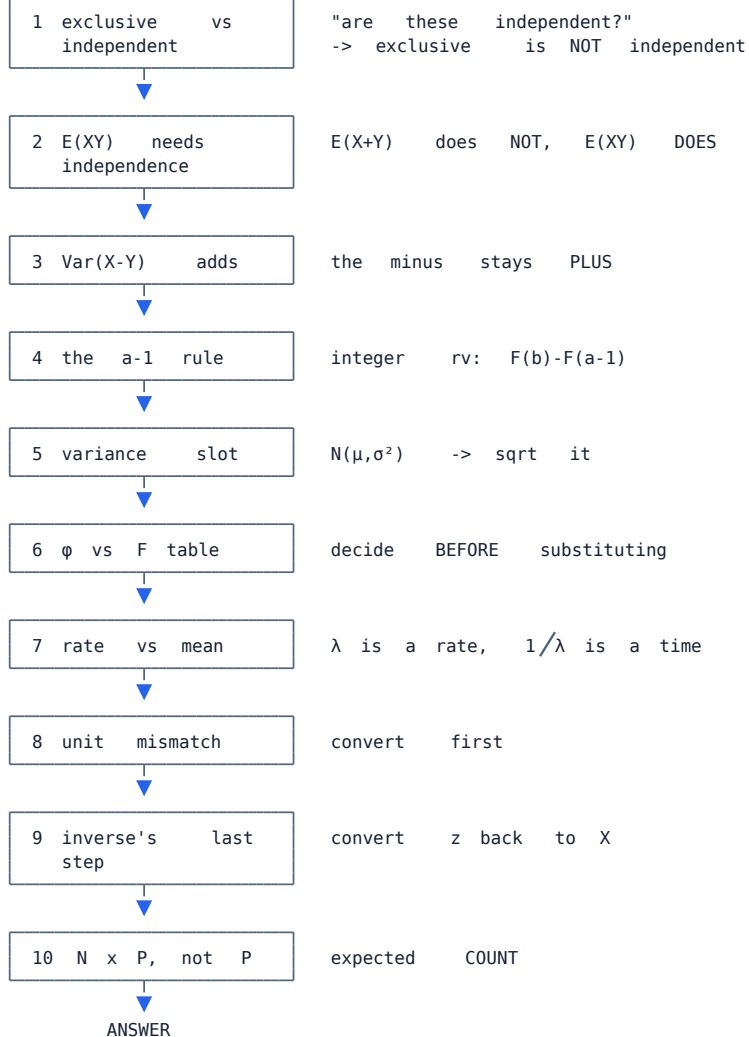


08 TRAPS: every trap and every source errata

THE TRAPS AS A GAUNTLET

a question walks through and each trap tries to catch it:

START →



THE ERRATA, BY WHERE THEY LIVE

SOURCE	ERRATA
ppt3 (binomial/Poiss)	E1 insurance 0.1745->0.1755
	E2 die count 0.549->0.5499
ppt4 (continuous)	E4 rate/mean wording p040
	E8 figure $\sigma=10$ -> use 5
notes deck	E13 bus integral bound 3 vs 2
clt deck	E6 impurity Z2 -0.4 -> -0.94
assignments	E3 0.0915->0.0916
	E5 second edition broken
	E9 0.5679->0.6225
	E10 mean 37.5->37.2
	E11 0.808->0.80
	E12 sd 0.975->0.97
	E14 0.8754->0.8753
	E16 tube 2/3 -> 8/27
	E17 row sums to 0.9
Chebyshev deck	E15 Q3 slide mis-stated
MTE schemes	E19 CDF option B -> D
	E20 infinite bounds -> 0..4
	E21 $E(t^2) \neq 0$ -> $E(t^2) \neq \theta^2$
OUR OWN mock	E18 "T4 smallest var" FALSE

THE 10-TRAP NIGHT LIST, AS A CHECKLIST

NIGHT BEFORE THE PAPER:

- ☐ 1 exclusive vs independent can state the difference in one line
- ☐ 2 $E(XY)$ needs independence vs $E(X+Y)$ always
- ☐ 3 $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$ the minus stays plus
- ☐ 4 $P(a \leq X \leq b) = F(b) - F(a-1)$ integer rv only
- ☐ 5 $N(\mu, \sigma^2)$ second slot is VAR sqrt before z
- ☐ 6 $\phi(z)$ area-from-0 vs $F(z)$ cum decide first
- ☐ 7 λ rate, $1/\lambda$ mean exponential
- ☐ 8 convert units FIRST 15/hr with 3 min
- ☐ 9 $X = \mu + z\sigma$ the inverse's last step
- ☐ 10 expected count = $N \times P$ not P

Two sections. First the concept traps (mistakes the QUESTIONS are built to catch). Then the source errata (numbers printed wrong in the slides and keys, with the correct value).

Part 1: concept traps

1	EXCLUSIVE vs INDEPENDENT. Mutually exclusive means no shared outcome. Independent means
$P(A \cap B) = P(A)P(B)$. Exclusive events with positive probability are NOT independent. This is	
the most common true/false the paper can carry.	
2	$E(XY) = E(X)E(Y)$ needs INDEPENDENCE. $E(X+Y) = E(X) + E(Y)$ does NOT. The paper probes exactly
this contrast.	
3	$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$ when independent. The minus stays plus. Squaring kills the sign.
4	THE a-1 RULE. For an integer rv, $P(a \leq X \leq b) = F(b) - F(a-1)$. Writing $F(a)$ is the slip. For
a continuous rv it does not matter.	
5	THE VARIANCE SLOT. $N(\mu, \sigma^2)$: the second number is the VARIANCE. Square-root it before
a z-score. "mean 70, variance 25" $\rightarrow \sigma = 5$.	
6	TWO TABLE CONVENTIONS. $\phi(z)$ is often the area from 0 to z; $F(z)$ is cumulative from $-\infty$.
MIXING them is the biggest normal error. Decide which the paper gives before substituting.	
7	RATE vs MEAN. Exponential: λ is a rate, $1/\lambda$ is the mean TIME. Getting the
reciprocal backwards changes the answer completely. (Slides get this wrong too, errata 4.)	
8	UNIT MISMATCH. A rate per hour with a time in minutes. Convert to one unit first.
9	THE INVERSE'S LAST STEP. Get z from the table, then convert $X = \mu + z \sigma$. Forgetting
the conversion leaves the answer in z-space.	
10	EXPECTED COUNT vs PROBABILITY. "Expected number out of N" is $N \times P$, not P. (0.5499 not
$5.5e-5$.)	
11	CHEBYSHEV k IN SIGMA UNITS. $k = \text{eps}/\sigma$. Raw units in, convert first. And the bound is
an UPPER bound on the tail, so report the exact value too when the distribution is known.	
12	CONSISTENCY vs SMALL VARIANCE. A fixed-n small variance is not consistency. Consistency is
about n growing. T1 with $\text{Var } 3\sigma^2$ at fixed $n=3$ is the trap.	
13	BIASED BEFORE VARIANCE. In "most efficient", a biased estimator is disqualified first,
even if its variance is smaller. Efficiency is among the UNBIASED only.	
14	EXACT vs ROUNDED. Carry exact values to the final line. Rounding mid-way and then rooting
or summing gives a slightly wrong number the key itself sometimes prints.	

1	EXCLUSIVE vs INDEPENDENT. Mutually exclusive means no shared outcome. Independent means
15	SUPPORT CHECK. A density is zero outside its support. Integrating over $(-\infty, \infty)$ when the
support is finite is a setup error (a scheme printed this, errata 20).	
16	VALID PDF CHECK. A pmf must sum to 1 over its support. A row that sums to 0.9 is not a pmf
(errata 17).	
17	"USE THE APPROXIMATION". When the paper says use the Poisson approximation to the
binomial, set $l = np$, not the exact binomial.	
18	LARGE SAMPLE LANGUAGE. "The mean of a sample of n " divides by the STANDARD ERROR, not by
sigma. A single observation uses sigma.	
19	CLT CONDITIONS. $n \geq 30$ is the course heuristic, not a universal theorem cutoff. For a
small non-normal population, do not assume the normal approximation without checking	
shape and tail behaviour.	
20	NORMAL POPULATION FALLBACK. A normal population is exactly normal at any n , so a small- n
question is fine if the population is stated normal. No large- n heuristic is needed.	

Part 2: source errata (the slides and keys print these wrong)

Use the computed value. Each entry is verified in the repo (reports/09-ERRATA.md).

E1	insurance Poisson: slide 0.1745, correct 0.1755	(ppt3 p027)
E2	irregular die count: slide 0.549, correct 0.5499	(ppt3 p018)
E3	assignment Q2 $P(X > 12)$: key 0.0915, correct 0.0916	(truncation)
E4	exponential rate/mean meaning inverted on p040, read p041 instead	
E5	assignment 1 long Q2 normalisation: the 12k version is a broken SECOND EDITION.	
batch-1 edition: $10k^2 + 9k = 1$, $k = 1/10$ exact, key checks out.		
second edition: $12k + 10k^2 = 1$, $k = 0.0782$, but its own key prints 1/10 which does		
NOT satisfy its own table. Do not drill the second edition.		
E6	impurity Z: slide prints $Z^2 = -0.4$, correct -0.94. Rounded table areas give	
0.1645; unrounded z values give about 0.1637. The slide's 0.1644 is also wrong.		
(clt p014)		
E7	$N(\mu, \sigma^2)$: the prose says variance, the notation says squared. Second slot IS	
variance.		

E1	insurance Poisson: slide 0.1745, correct 0.1755	(ppt3 p027)
E8	ppt4 p030 figure labels $\sigma = 10$ under $\mu = 8$, but the arithmetic uses $\sigma = 5$.	
Use 5. (errata 8)		
E9	assignment 2 B Q5: key prints 0.5679, correct 0.6225	(exponential conditional)
E10	assignment 2 C Q2: key prints mean 37.5, correct 37.2 (the key's own cutoff agrees	
with 37.2). Use $\mu = 37.2$, $\sigma = 28.2$, cutoff 30.4.		
E11	assignment 1 long Q3: key prints 0.808, both cases are exactly 0.80 (hypergeometric	
mean equals binomial mean here)		
E12	assignment 1 app Q3: key rounds twice, sd 0.975 $\rightarrow$ 0.98. Exact is 0.9734 $\rightarrow$ 0.97.	
E13	deck bus example p127/p129: the printed first integral bound (3) contradicts the stated	
event ($Y < 2$). Final answer 2/5 is correct for " $Y < 2$ or $Y > 6$ ". Use $P(Y < 2) + P(Y > 6) = 0.08 + 0.32 = 0.40$		
E14	assignment 2 C Q3: key 0.8754, exact 0.8753.	
E15	Chebyshev L10-11 Q3 slide: states one row, works another. It also writes a reciprocal	
template. USE Q1 ($E = 3$, $E(X^2) = 13$, bound 21/25) AND Q2 ($\mu = 10$, $\text{var} = 4$) ONLY. Do not quote		
Q3's row. (Full retraction detail in the repo errata, incl. two claims that were		
themselves retracted after checking.)		
E16	tube question (2024 A1 Q12 and 2025-26 Q14): key part (ii) prints 2/3, correct is the	
probability that ALL THREE survive = $(2/3)^3 = 8/27$. The key carried the single-tube		
number into the three-tube part. Parts (i) 1/27, (iii) 0.25, (iv) 1.7 are correct.		
E17	2024-25 assignment 1 Q16: the probability row sums to 0.9, not 1. It is not a valid pmf.	
State the renormalisation you choose (divide by 0.9) or ask for the correction.		
E18	mock paper B4 (our own): the Note says T4 has the smallest variance. False, T4 is third	
of four (order T3, T2, T4, T1). T4 is biased, keep that part, drop the superlative.		
E19	2024 MTE QA2 CDF: scheme marks B, correct is option D (the CDF must accumulate the mass	
from 0 to 1). Option B gives $F(1) = 1.5$, impossible.		
E20	2025 MTE scheme Q4: displays infinite integration bounds, but the support is $0 < x < 4$ and	
the arithmetic uses 0 and 4. The constants are right, the printed bounds are the slip.		
E21	2025 MTE scheme Q8(ii): prints $E(t^2) \neq 0$, correct is $E(t^2) = \theta^2$.	

The 10-item trap list to read the night before

1	exclusive vs independent
2	$E(XY)$ needs independence, $E(X+Y)$ does not
3	$\text{Var}(X-Y)$ adds
4	the $a-1$ in $F(b)-F(a-1)$
5	variance slot in $N(\mu, \sigma^2)$, sqrt it
6	ϕ vs F table convention
7	rate vs mean in the exponential
8	unit mismatch
9	the inverse's last conversion step
10	expected count = $N \times P$, not P